Prof. Barry Quinn

Professor of Finance and Financial Technology

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Personal Statement

Professor Barry Quinn is Professor of Finance & Financial Technology at Ulster University Business School and Director of the Centre for Finance and Responsible Technology. The Centre is guided by three pillars: statistical rigour and intellectual humility; practical industry relevance; and responsible algorithmic innovation.

Barry is a Chartered Statistician and views econometrics as an applied statistical science. His work combines applied econometrics and causal inference with trustworthy machine learning and computational approaches to produce decision-relevant evidence for markets, risk, market integrity, and regulatory compliance. In practice, that means making assumptions explicit, quantifying uncertainty, stress-testing results, and being clear about limits—so we can say plainly when evidence does (and does not) support confident conclusions.

His research spans causal policy evaluation, algorithmic trading risk detection, and trustworthy AI methods, and is collaborative and interdisciplinary—bridging finance, statistics, computer science, and law.

His research sits at the intersection of finance, statistical learning, and responsible use of algorithms, with a focus on markets and institutions, risk measurement, and evidence for regulation and compliance. He works collaboratively across disciplines and with external partners, aiming to reduce avoidable complexity and produce analysis that stands up under operational and regulatory scrutiny.

Before entering academia, Barry worked in financial markets, specialising in currency trading and liquidity management. He holds a PhD in Finance and an MSc in Artificial Intelligence from Queen's University Belfast. He teaches across quantitative finance, econometrics, and AI applications in finance, emphasising ethical practice, reproducibility, and critical thinking.

Education and Professional Qualifications

Qualification	Institution	Year
B.Sc.(Hons) Accounting and Finance	Queen's University Belfast	1995
MSc Quantitative Finance	RMIT University Melbourne	2006
Ph.D. Finance	Queen's University Belfast	2012
Chartered Statistician	Royal Statistical Society	2019
Advanced Data Science Professional	Royal Statistical Society	2023
MSc Artificial Intelligence	Queen's University Belfast	2025

Teaching

I prepare students to reason carefully about finance and technology—teaching them to make assumptions explicit, quantify uncertainty, and be honest about what evidence does and does not support. My goal is to develop analysts who are technically capable, intellectually honest, and able to produce work that is transparent and reproducible. These qualities—rigour, humility, and critical thinking—equip students for roles where decisions carry real consequences.

I have extensive teaching experience at both undergraduate and postgraduate levels, specialising in:

1. **Algorithmic Trading and Investment**: A postgraduate course grounded in the López de Prado framework for machine learning in financial markets. Covers the full pipeline from financial data structures and feature engineering to portfolio construction, backtesting, and strategy evaluation—with emphasis on avoiding the methodological pitfalls (backtest overfitting, data snooping, and leakage) that plague practitioner and academic research alike.

2. **Financial Econometrics**: Taught as an applied statistical science rather than a recipe book. Students learn to treat econometric models as tools for quantifying uncertainty and stress-testing assumptions—not as black boxes that produce publishable numbers. The course confronts the replication crisis and the statistical practices that drive it, asking students to be explicit about what their models can and cannot establish.

3. **FinTech and Data Science** (Ulster University): The disciplined study of variation and uncertainty in financial data—covering return predictability, volatility modelling, factor models, and ML for cross-sectional analysis. Emphasises distinguishing signal from noise, proper model validation, and intellectual humility about the limits of prediction in noisy markets. Supported by an open companion textbook, *Statistical Science for Finance: Rigorous Methods for Technology-Enabled Markets*.
4. **MiniMBA in FinTech and Responsible Data Science** (Ulster University, Magee Campus, 2026–): A DfE-sponsored, credit-bearing four-day intensive (one day delivered online) at UUBS/Magee, structured around three pillars that mirror the Centre for Finance and Responsible Technology's mission: the economics of FinTech (cost puzzle, platforms, prediction economics); responsible data science (validation discipline, overfitting, walk-forward testing, evidence standards); and AI in finance with a regulatory compliance focus, drawing directly on CFRT/UKFin+ research into governance and graduated automation. Materials are published openly at quinfer.github.io/minimba with interactive Google Colab labs. The inaugural 2026 cohort produced a Compare NI FinTech Scholarship winner. Next delivery: Spring 2027.
5. **Trading Principles** (Queen's University Belfast): A postgraduate learning-by-doing course in market microstructure, built around approximately 18 live in-class trading simulations using UpTick technology integrated with Bloomberg Terminal data. Students assume distinct industry roles—market maker, speculator, liquidity trader—and their collective decisions endogenously determine prices and liquidity, making core concepts such as the law of one price, price formation, market efficiency, and event arbitrage directly observable consequences of their own behaviour. Simulation performance counts toward the final grade, removing the gap between doing and understanding. The course pairs foundational academic work (Kyle 1985; Glosten-Milgrom 1985) with industry research to bridge theory and practice.

This commitment to experiential learning extends beyond the classroom. At Queen's I co-founded the Queen's Student Managed Fund (2012–2024), giving students stewardship of a real investment portfolio. At Ulster I am establishing a new Trading and Investment Group, funded by a £100K alumni philanthropic donation in partnership with Innovation Ulster Ltd, launching September 2026—bringing the same learning-by-doing philosophy to a new institutional context.

Teaching Publications

- Quinn, Barry (2026). *Statistical Science for Finance: Rigorous Methods for Technology-Enabled Markets*. Open educational resource and companion textbook for FIN306, Ulster University Business School. https://quinfer.github.io/financial-data-science/
- Quinn, Barry, Hanna, Alan, Gallagher, Aine (2018). Queen's Student Managed Fund: Investing in the Student Experience. *Reflections*, 27, pp. 16–17.

Research

I have published fifteen peer-reviewed academic papers (1 ABS4, 11 ABS3s, and 3 ABS2s), three commissioned research reports, and two statistical software packages. My research sits at the intersection of finance, statistical learning, and responsible use of algorithms—with a focus on markets and institutions, risk measurement, and evidence for regulation and compliance. The work is collaborative, crossing disciplines and industry boundaries, and aims to reduce avoidable complexity: producing analysis that is transparent, reproducible, and stands up under operational and regulatory scrutiny.

Research Expertise

Three methodological strands recur across my programme. The first is **trustworthy machine learning for market integrity and risk**. With colleagues at Queen's University Belfast I have developed representation-learning methods for detecting market manipulation in high-frequency equity data — a dual-branch self-supervised framework combining frequency-informed anomaly synthesis with domain-specific features (published online June 2026 in *Information Processing & Management*), a related inexact-supervised time-series paper at revise-and-resubmit at *IEEE Transactions on Cybernetics* (FMISD), and a signal-amplification technique for rare-event manipulation detection (Dai, Quinn, Kearney & Wang, retargeting *Quantitative Finance* after desk rejection at *Economics Letters*). Related work applies transformer models to sentiment-conditioned crash-risk modelling (Birem, Abidi Perier, Quinn & Kearney, resubmitting to *European Financial Management* after rejection at the *European Journal of Operational Research*). The methodological commitment running through this strand is that model performance and model trustworthiness are separable properties, and both must be evidenced before machine learning is deployed in operational or regulatory settings.

The second strand is **computer vision for socio-economic measurement** — using vision models on real-world imagery to produce auditable indicators of phenomena that are otherwise hard to quantify consistently. This includes the Hannon, French, Quinn & O'Hagan geospatial study of vehicle crime in Northern Ireland (rejected at *Insurance: Mathematics and Economics*; being resubmitted to *Regional Science and Urban Economics*), which extracts environmental features from street-level imagery; Cochrane, Bryan, Quinn, Graham & French's IGPSJ Working Paper (WP 02/26, 2026) on spatial correlates of flag displays across Northern Ireland; and the hierarchical flag-classification programme developed during my MSc AI dissertation at Queen's (4,501-image Northern Ireland dataset; 70→16→7 hierarchy-aware taxonomy; ViT-H-14 baseline at 94.78% on the 7-category task), currently in manuscript preparation. Both projects treat vision pipelines as instruments of measurement, and are evaluated on measurement-science grounds rather than on benchmark accuracy alone.

The third strand is **causal and statistical evidence for governance and regulation**. This covers the Kearney, Quinn & Pramanick study of innovation signals and strategic exits in computational approaches to financial regulation (desk rejected at the *Journal of Business Venturing*, retargeting); the Quinn, Sheenan & Zhang analysis of Basel III's cross-border transmission through foreign bank subsidiaries (Centre for Responsible Banking & Finance Working Paper 25-017; *Journal of Financial Stability* submission in preparation); and the collaborative doctoral partnerships with Pytillia, Napier AI and Funds Axis that apply causal-inference and econometric methods to live regulatory and compliance problems. In this strand the methodological priority is that evidence survive operational and regulatory scrutiny — assumptions explicit, uncertainty quantified, and claims proportionate to what the data can support.

Peer-Reviewed Publications

- Dai, Yongsheng, Quinn, Barry, Kearney, Fearghal, Liu, Weilong, Spence, Ivor, Rafferty, Karen, Wang, Hui (2026). Detecting market manipulation with dual-branch self-supervised learning: A unified framework integrating frequency-informed anomaly synthesis and domain-specific features. *Information Processing & Management* 63(8), Article 104961, 1–30 (ABS2; published online 8 Jun 2026; Pure)
- Wang, Hui, Dai, Yongsheng, Spence, Ivor, Rafferty, Karen, Quinn, Barry, Huang, Ji (2025). TDSRL: Time Series Dual Self-Supervised Representation Learning for Anomaly Detection from Different Perspectives. IEEE Internet of Things Journal
- Peng, Qiao (Olivia), McKillop, Donal, Quinn, Barry, Liu, Kailong (2025). Modeling and predicting failure in US credit unions. *International Journal of Forecasting* 41(3), 1237–1259 (ABS3; published 4 Jun 2025; Pure)
- Liu, Weilong, Zhang, Yong, Liu, Kailong, Quinn, Barry, Yang, Xingyu, Peng, Qiao (2024). Evolutionary Multi-Objective Optimisation for Large-Scale Portfolio Selection With Both Random and Uncertain Returns. *IEEE Transactions on Evolutionary Computation* (ABS4)
- Bouri, E., Quinn, B., Sheenan, L. & Tang, Y. (2024). Investigating extreme linkage topology in the aerospace and defence industry. *International Review of Financial Analysis* (ABS3)
- Quinn, Barry, Gallagher, Ronan, Kuosmanen, Timo (2023). Lurking in the shadows: The impact of CO2 emissions target setting on carbon pricing in the Kyoto agreement period. *Energy Economics* (ABS3)

- Liu, Jiadong, Papailias, Fotis, Quinn, Barry (2021). Direction-of-change forecasting in commodity futures markets. *International Review of Financial Analysis* (ABS3)
- Gallagher, Ronan, Quinn, Barry (2020). Regulatory own goals: The unintended consequences of economic regulation in professional football. *European Sport Management Quarterly* (ABS3)
- McKillop, Donal, French, Declan, Quinn, Barry, Sobiech, Anna L, Wilson, John OS (2020). Cooperative financial institutions: A review of the literature. *International Review of Financial Analysis* (ABS3)
- Quinn, Barry, Hanna, Alan, MacDonald, Fred (2018). Picking up the pennies in front of the bulldozer: The profitability of gilt based trading strategies. *Finance Research Letters* (ABS2)
- McKillop, Donal G, Quinn, Barry (2017). Irish credit unions: Differential regulation based on business model complexity. *The British Accounting Review* (ABS3)
- Ayadi, Rym, Naceur, Sami Ben, Casu, Barbara, Quinn, Barry (2016). Does Basel compliance matter for bank performance? *Journal of Financial Stability* (ABS3)
- Glass, J, McKillop, Donal, Quinn, Barry (2015). Modelling the Performance of Irish Credit Unions, 2002-2010. *Financial Accountability & Management* (ABS3)
- McKillop, Donal G, Quinn, Barry (2015). Web adoption by Irish credit unions: Performance implications. *Annals of Public and Cooperative Economics* (ABS2)
- Glass, J Colin, McKillop, Donal G, Quinn, Barry, Wilson, John (2014). Cooperative bank efficiency in Japan: a parametric distance function analysis. *The European Journal of Finance* (ABS3)
- Quinn, Barry, McKillop, Donal (2009). Internet banking and Irish credit unions. *International Journal of Cooperative Management* (ABS2)
- Quinn, Barry, McKillop, Donal (2009). Cost performance of Irish credit unions. *Journal of Cooperative Studies* (ABS2)

Software Publications

Table 2: Software Publications

Package	Year	Description
Time Series Econometrics (tsfe) (DOI: 10.5281/zenodo.6376113)	2022	R Package for time series analysis
Financial Machine Learning (fml) (DOI: 10.5281/zenodo.6376114)	2020	R Package for financial ML

Working Papers

- Birem, AbderRaouf, Abidi Perier, Zineb, Quinn, Barry, Kearney, Fearghal (Resubmitting). *Transformer-Based Sentiment Analysis for Stock Market Crash Risk*. Rejected from *European Journal of Operational Research* (manuscript EJOR-S-26-00902); being resubmitted to *European Financial Management* (ABS 3).
- Dai, Yongsheng, Quinn, Barry, Kearney, Fearghal, Wang, Hui (Resubmitting). *Signal Amplification and Strategic Deterrence in Market Surveillance* (overhaul of, and formerly titled, *Amplifying Market Manipulation Detection Signals*). Desk rejected at *Economics Letters* (EL66561); rebuilt as a theory-forward article (Fisher–Mahalanobis amplification theorem plus a strategic-deterrence game and Shenzhen Level-2 empirics) and being targeted at *Quantitative Finance* (ABS 3). Materials: `Research/signal-amplification-theory-full/`.
- Dai, Yongsheng, Huang, Ji, Ren, Spence, Ivor, Rafferty, Karen, Quinn, Barry, Wang, Hui (Revise and resubmit). *FMISD: Fine-grained Multi-Scale Modeling for Inexact Supervised Anomaly*

Detection of Time Series. Revise-and-resubmit at *IEEE Transactions on Cybernetics* (ABS 3). Materials: `Research/submissions-to-journals/market-manipulation-papers/fmisd-ieee-trans-cybernetics/submission-history/`.

- Quinn, Barry, Sheenan, Lisa (UCD), Zhang, Ying (Veronica) (2026). *Regulatory Reach or Regulatory Arbitrage? Basel III's Cross-Border Transmission Through Foreign Bank Subsidiaries* (earlier titles: *Basel III's Impact on Chinese Banking Risk-Taking*; *Ownership Dynamics, Risk and Regulation in Chinese Banking: New Evidence*). Manuscript complete; accepted to and presented by Ying (Veronica) Zhang at the Irish Academy of Finance (IAF) 7th Annual Conference, 21–22 May 2026, The Johnstown Estate Hotel, Enfield, Co. Meath (submission ID 49); submission to *Journal of Financial Stability* (ABS 3) planned before the end of summer 2026. Centre for Responsible Banking & Finance Working Paper Series, WP No. 25-017. Latest draft: `Research/basel-china-banks/final_submission/`.
  - *Note:* CRBF WP 25-017 was issued under an earlier title and is cited as originally published; the live manuscript has since been retitled as above.
- Kearney, Fearghal, Quinn, Barry, Pramanick, Abhishek (Desk rejected at *Journal of Business Venturing*, 2026; retargeting). *Innovation Signals and Strategic Exits: How Technological Readiness Shapes Outcomes in Computational Approaches to Financial Regulation* (manifest title: *Modelling and Forecasting the Evolution of RegTech*). Materials: `Research/regtech-survival-jbv-desk-reject/`.
- Hannon, James (Corresponding Author), French, Declan, Quinn, Barry, O'Hagan, Adrian (Resubmitting). *Geospatial modeling of vehicle crime in Northern Ireland using computer vision to identify environmental factors*. Rejected from *Insurance: Mathematics and Economics* (IME-D-25-00419); being resubmitted to *Regional Science and Urban Economics* (ABS 3).
- Cochrane, Brandon, Bryan, Dominic, Quinn, Barry, Graham, Bryon, French, Declan (Corresponding Author) (2026). *Flying the Flag: Cultural Expression as Territorial Demarcation*. Institute for Global Peace, Security and Justice Working Paper Series, WP 02/26, Queen's University Belfast. ISSN 2399-5130. https://www.qub.ac.uk/sites/institute-for-global-peace-security-justice/FileStore/WP-02-26.pdf. Materials: `Research/flags project papers/01-working-papers/brandon-phd-chapter/working-paper/IGPSJ-WP-02-26/`.
- Campbell, Gareth, Kelly, Colm, Quinn, Barry. *When Simplicity Beats Sophistication: Market Size and the Limits of Optimal Factor Construction*. Working paper; Asset Pricing Trees on UK equities, 1869–2024. Target *Review of Finance* (ABS 4) `[verify - RoF/JoF target discussed with co-authors]`. Materials: `Research/machine-learning-asset-pricing/`.
- Quinn, Barry. *Sensitivity Analysis and Partial Identification for Covariate Effects in Structural Topic Models*. Working paper (methods). Journal target `[verify - JBF/EJF, or Review of Finance/JFQA if framed around the price of risk]`. Materials: `Research/stm_critique/`.
- Quinn, Barry, Gallagher, Ronan. *Great Expectations: Relative Performance and Managerial Turnover in Professional Football*. Drafting; near submission. Reframed as a contextual stochastic optimisation / operational-research paper (dynamic horizon-dependent evaluation policy, dual-LLM turnover classification, ~7,132 spells). Target *European Journal of Operational Research* (ABS 4). Materials: `Research/great-expectations/`.
- Quinn, Barry (2025). *Hierarchical Flag Classification through Economic Domain Knowledge: A Vision Transformer Approach for Cultural Symbol Recognition*. MSc AI thesis and reproducible research codebase (economic-flag-classification), Queen's University Belfast. Manuscript in preparation; being reframed for *Urban Studies* (ABS 3) — working title *Territorial markers at street level: measuring contested symbolism in Northern Ireland with Google Street View and a theory-derived vision classifier* — superseding the earlier *Decision Support Systems* framing. Reframe pack: `economic-flag-classification/literature/urban_studies_reframe/`.
- Quinn, Barry, Wang, Ying (Veronica). *The Efficiency Returns to Cloud Computing: Evidence from Staggered Data Protection Laws*. Empirical measurement work stage (moved on from the earlier research-proposal stage). Journal target `[verify]`. Materials: `Research/cloud-computing-paradox/`.

- Quinn, Barry, Gallagher, Ronan, Birem, AbderRaouf. *Board Composition and M&A Announcement Returns*. Early stage (drafting). Journal target `[verify]`. Materials: `Research/computational-governance-lab/`.
- Quinn, Barry, Peng, Qiao (Olivia), Liu, Weilong. *Climate Risk and Credit Union Lending: Channel-Specific State Heterogeneity*. First draft complete; revising toward submission to *Journal of Financial Stability* (ABS 3) over summer 2026 (the noted natural home; branch-level identification being strengthened to meet JFS referees). Materials: `Research/credit-unions-climate-risk/`.
- Walker, Clive, Quinn, Barry. *Do Exogenous Shocks Reshape Financial Narratives?*. Drafting. Text-as-data / narrative-shocks study on the FT leaders corpus (1888–2017). Target ladder `[verify]`: *Journal of Banking & Finance* / *European Journal of Finance* (regime-shift framing), or *Review of Finance* / *JFQA* if the price-of-risk link is established. Materials: `Research/stm_critique/` (manifest `PROJECT-ft-narrative-shocks.yaml`). Note: distinct from the STM methods paper above; folder consolidation pending the "one vs two papers" decision with Clive Walker.
- Quinn, Barry `[verify - co-authors]`. *Adaptive Capital and Patent Boundary Indeterminacy*. Drafting (theory paper; Route B of a planned two-paper sequence). Target *Industrial and Corporate Change* (ABS 3) or *Technological Forecasting & Social Change* (ABS 3); *Research Policy* (ABS 4*) a long shot for pure theory. Materials: `Research/AI and Adaptive Capital Idea/` (no `PROJECT.yaml` yet).
- Quinn, Barry `[verify - co-authors]`. *Beyond Algorithms*. Drafting (empirical paper; Route A, planned as a follow-up to the theory paper). PATSTAT-based test of the adaptive-capital predictions. Target *Research Policy* (ABS 4*, verified against the AJG 2024 master), with *RAND Journal of Economics* (ABS 4) or *Industrial and Corporate Change* (ABS 3) as alternatives. Materials: `Research/AI and Adaptive Capital Idea/`.

Research Reports

- Bryon Graham, Barry Quinn (2019). *Price comparison and web analytics* (commissioned by InvestNI).
- Bryon Graham, Barry Quinn (2017). *Machine learning and predictive analytics in the mall retail business* (commissioned by InvestNI).
- Fearghal Kearney, Barry Quinn (2020). *The theoretical foundations of value at risk modelling* (commissioned by InvestNI and Funds Axis Ltd).
- French, Declan, McKillop, Donal, Quinn, Barry (2018). *Landscape review of Northern Ireland Credit Union Sector* (commissioned by the NI Department of Communities and Rural Affairs).
- McKillop, Donal, Quinn, Barry (2012). *Report of the Irish Commission on Credit Unions — An Efficiency Study* (4* REF 2014 Impact Case).

Research Impact and Knowledge Transfer

1. *Three PhD Scholarships — Centre for Finance and Responsible Technology* (Department of the Economy NI, £360K, 2025), including two collaborative doctoral partnerships with local firms: Pytillia and Napier AI.
2. *Understanding and Enhancing regulatory compliance in the Investment Management industry using AI* with Funds Axis Ltd and Momentum 1.0 (UKFin+ / UKRI, November 2024 – November 2025).
3. *Anomaly detection of large heterogeneous trading transaction and communication data* with Citigroup Belfast and Momentum 1.0 (2022–2023).
4. *AI and Advanced Retail Analytics* with Pearlai Ltd and Dr Byron Graham (KTP, 2017–2019).
5. *E.S.G. Fair Value Analytics* with Research Associates Dr Lisa Sheenan and Dr Byron Graham (KTP, 2021).
6. *Tail Risk Analytics, Stress Testing and Scenario Analytics* with Funds Axis Ltd, Dr Fearghal Kearney and Dr Colm Kelly (KTP, 2021–2023).
7. *Financial Conduct Authority Tech Sprint Mentor* (2023–present).
8. *AI and the Future of Financial Regulation* with Fearghal Kearney and Abhishek Pramanick, published in The Economic Observatory (2023).

9. *Hierarchical Flag Classification for Cultural-Symbol Monitoring* (QUB MSc AI project, 2024–2025): developed a 4,501-image Northern Ireland street-scene dataset and a hierarchy-aware taxonomy (70→16→7) with a ViT-H-14 baseline achieving 94.78% on the 7-category task; code and methods are documented in economic-flag-classification.
10. *AI Adoption, Opportunities & Impact for SMEs* (Intertrade Ireland Tender Scheme; UU Worktribe 924736; PI Barry Quinn): funded research on all-island SME AI adoption, opportunities and impact; InterTradeIreland research report in preparation; academic publication under consideration.
11. *Trading and Investment Group, Ulster University Business School* (£100K alumni philanthropic donation, 2026; in partnership with Innovation Ulster Ltd): establishing a student-run trading and investment group at Ulster University Business School, providing experiential learning in live markets and real portfolio management; launching September 2026.
12. *MiniMBA in FinTech and Responsible Data Science* (DfE-sponsored, Magee Campus, Ulster University Business School, 2026): inaugural cohort of a credit-bearing four-day intensive (one day online) executive education programme covering the economics of FinTech, responsible data science and validation discipline, and AI in finance/regulation — with Day 3 drawing directly on CFRT/UKFin+ research. Materials published openly at quinfer.github.io/minimba. A student from the cohort was awarded the Compare NI FinTech Scholarship. Programme will run again Spring 2027.

Completed PhD Supervisions

1. Dr Jiadong Liu (graduated 2018): *Momentum in Empirical Asset Pricing* **(Principal Supervisor)**.
2. Dr Ashleigh Neil (graduated 2019): *Law and Financial Stability* **(Principal Supervisor)**.
3. Dr Colm Kelly (graduated 2021): *Machine Learning in Empirical Asset Pricing* **(Principal Supervisor)**.
4. Veronica Zhang (2024): *Capital Policy and State Sponsorship in Chinese Banking* **(Principal Supervisor)**.
5. Dr Kevin Johnson (graduated 2019): *A Case Study in Industrial Consolidation — The Irish Credit Union Sector* **(Co-Supervisor)**.
6. Dr Qiao (Olivia) Peng (graduated 2021): *US Credit Union Mergers — Causes and Consequences* **(Co-Supervisor)**.

Current PhD Supervisions

1. Fei Li: *Sustainable Investing, Reporting and Gender Diversity* (fully-funded QBS scholarship) **(Principal Supervisor)**.
2. Abhishek Pramanick: *AI and the Future of Financial Regulation* (fully-funded QBS scholarship) **(Principal Supervisor)**.
3. Dongwei (Jeffrey) Li: *The Economics of AI Patents and Productivity in Finance* (fully-funded DofE scholarship) **(Principal Supervisor)**.
4. Yuqi Ding: *The Economics of Generative AI in Finance* (fully-funded QBS scholarship) **(Principal Supervisor)**.
5. Yongsheng Dai: *Market Manipulation and Financial Anomaly Detection* (fully-funded PWC scholarship) **(Co-Supervisor)**.
6. Brandon Cochrane: *Economic Cost of Cultural Displays in Northern Ireland* (fully-funded DofE CAST scholarship) **(Co-Supervisor)**.

Visiting PhD Supervision

1. Weilong Liu (2022; Lingnan University College of Sun Yat-sen University, Guangzhou): *Market Manipulation and AI Anomaly Detection*.
2. James Hannon (2023; University College Dublin): *Leveraging Computer Vision to Understand and Enhance Insurance Pricing*.
3. Birem Abderraouf (2025; UPEC Paris): *Generative AI in Capital Markets*.

Conference Presentations

- Invited Speaker, *Dynamic Signal Weighting of Performance Signals for Managerial Turnover Decisions: Evidence from Professional Football*, INFINIT Conference, Edinburgh Napier University, Jun 2025.
- Invited Talk, *Rethinking Research Impact: Combining Effect Sizes and Economic Significance in Finance*, Royal Statistical Society Annual Conference, Brighton UK, Sep 2024.
- Keynote Speaker, *Estimating Systemic Risk*, Irish Finance Association, Maynooth University, Ireland, April 2023.
- Invited Talk, *Teaching Data Science in the Age of FinTech*, Royal Statistical Society Annual Conference, Aberdeen UK, Sep 2022.
- Invited Talk, *Carbon Pricing and Machine Learning*, Multidisciplinary Workshop on Fintech, Islamic Finance and Sustainability (online), Hamad Bin Khalifa University, Qatar, Nov 2022.
- Invited Speaker, *Understanding Fintech and Financial Stability*, International Workshop on Financial System Architecture and Stability, Bayes Business School, London, Sep 2018.
- Invited Speaker, *Systemic Risk and Basel Compliance*, British Accounting and Finance Association Annual Conference, London, Apr 2018.
- Invited Speaker, *Differential Regulation of Irish Credit Unions: Does One Size Fit All?*, 2nd Conference on Contemporary Issues in Banking, Centre for Responsible Banking and Finance, St Andrews, Dec 2017.
- Invited Speaker, *Business Model Diversity, Efficiency and Productivity of Cooperatives*, European Workshop in Efficiency and Productivity Analysis, Aalto University, Finland, Jun 2017.
- Invited Panellist, *Statistical Inference and Credibility in Finance*, Emerging Scholars in Banking and Finance, Bayes Business School, London, Dec 2016.
- Invited Speaker, *Capital Regulation Compliance and the Performance of European Banks*, International Workshop on Financial System Architecture and Stability, HEC Montreal, Aug 2016.
- Participant, *Bloomberg Annual Educational Symposium*, Bloomberg London HQ, Sep 2015.

Research Grant Income

Table 3: Research Grant Income

Source	Role	Start	End	People	Title	Full Economic Costs (£)	Type
(Department of the Economy NI)	PI	2025	NA	Pytillia Napier AI	Three PhD scholarships to support the Centre for Finance and Responsible Technology (including two collaborative doctoral partnerships)	£360,000	Research
(UK-Fin+/EPSRC)	PI	2024	2025	Jesus Del Rincon Martinez Abhishek Pramanick Darren Burrows (CEO of Funds Axis)	Leveraging AI to understand and improve regulatory compliance in the Investment Management Industry	£100,000	Research
DofE CAST Award NI	Co-I	2023	NA	Declan French	Exploring the Welfare Cost of Cultural Displays in Northern Ireland using multimodal Generative AI	£70,000	Research

(continued)

Source	Role	Start	End	People	Title	Full Economic Costs (£)	Type
Innovate UK	PI	2022	2023	Lisa Sheenan Byron Graham	E.S.G fair value analytics platform: using state-of-the-art financial data science and business analytics to design a fair-value ESG prediction engine	£173,000	Research
Innovate UK	PI	2021	2023	Fearghal, Kearney	Regulatory technology and portfolio analytics using state-of-the-art econometrics and financial machine learning	£173,000	Research
Innovate UK	PI	2018	2021	Byron, Graham	Designing and deploying a retail analytics platform using advanced analytics and machine learning	£165,000	Research
DofE CAST Award NI	Co-I	2021	NA	Donal McKillop	PhD investigation into how AI innovation affect a highly valued financial service provision experience	£70,000	Research
Phoenix Natural Gas Ltd	PI	2014	2015	Alan Hanna Fotis Papailias	Forecasting daily demand for natural gas in Northern Ireland	£15,000	Consulting
DofE	Co-I	2016	2018	Donal McKillop Declan French	Landscape review of Northern Ireland Credit Unions	£32,000	Consulting
InvestNI	Co-I	2018	NA	Byron Graham	Review of state-of-the-art machine learning in the context of retail analytics and monitoring	£5,000	Consulting
InvestNI	Co-I	2019	2019	Byron Graham	Review of AI and web analytics for price comparison	£5,000	Consulting
InvestNI	Co-I	2021	2021	Fearghal Kearney	Review of Value at Risk analytics at the intersection of econometrics and machine learning	£5,000	Consulting
QBS CEBR	PI	2016	2019	Institute of Directors	Estimating the costs and benefits of corporation tax policy change in Northern Ireland	£3,500	Research

Total Research Income: approximately £1,176,500 in direct awards as PI or Co-I across 13 projects (source: `data/Grant_income.csv`).

Citizenship

Academic Service

- **Director**, Centre for Finance and Responsible Technology, Ulster University Business School, September 2025–.

Table 4: Professional Experience

Period	Position	Institution
2025 - Present	Director, Centre for Finance and Responsible Technology (from September 2025)	Ulster University Business School
2024 - Present	Professor of Finance and Financial Technology	Ulster University Business School
2020 - 2024	Senior Lecturer in Finance, Technology and Data Science	Queen's Management School
2010 - 2020	Lecturer	Department of Finance, Queen's Management School
2009 - 2009	Teaching Fellow	Business School, Ulster University
2005 - 2005	Teaching Fellow and Quantitative Researcher	Department of Finance, RMIT University Melbourne
1998 - 2004	Currency Trader and Liquidity Manager	Janus Henderson Investors, London
1995 - 1998	Financial Adviser	City Financial Partners, London

- **Founder**, Trading and Investment Group, Ulster University Business School, 2026– (£100K alumni philanthropic fund; in partnership with Innovation Ulster Ltd; launching September 2026).
- **Programme Lead**, MiniMBA in FinTech and Responsible Data Science, Magee Campus, Ulster University Business School, 2026– (DfE-sponsored; credit-bearing four-day intensive (one day online); inaugural cohort 2026; Compare NI FinTech Scholarship winner from cohort; open materials at quinfer.github.io/minimba; next delivery Spring 2027).
- **Co-founder**, Finance and AI Research Lab (with Dr Fearghal Kearney), Queen's University Belfast, 2022–2024.
- **Lead Developer**, Queen's Business School Remote Analytics Lab Platform (Q-RaP), 2021–2024.
- **QUB Academic Lead**, Steering Group for Northern Ireland Global Centre in Secure Connected Intelligence for Regulatory Technology in Finance, 2022–2024.
- **Programme Director**, MSc Quantitative Finance, Queen's University Belfast, 2018–2022.
- **Senior Research Fellow** (Sabbatical), International Research Centre on Cooperative Finance, HEC Montreal University, Canada, May–September 2018.
- **Programme Director**, MSc Computational Finance & Trading, Queen's University Belfast, 2014–2018.
- **Co-founder**, Queen's Student Managed Fund, 2012–2024.
- **Implementation Team Member**, Bloomberg Trading Room, Queen's University Belfast.
- **External Grant Reviewer**, UKFin+ (UKRI specialist funded project) — reviewed feasibility grant application (£100K) for financial services innovation research, 2024.
- **Organiser**, Symposium on AI and the Future of Financial Regulation, Queen's University Belfast, November 2023.
- **Organiser**, Symposium on Safety and Assurance in Finance, Queen's Business School, November 2024.
- **Chair**, Panel Debate on "Digital Regulation: Shaping Digital Markets and Safeguarding Consumer Rights in Northern Ireland", Northern Ireland Competition Forum, May 2024.

Professional Accreditations

- Chartered Statistician, Royal Statistical Society (2019).
- Advanced Data Science Professional, Royal Statistical Society (2023).

Awards and Recognition

- Teaching Fellow of the Higher Education Authority (2012).
- QUB Teaching Award (2016).
- 1st place, CFA European Quantitative Finance Awards (2018).
- EEECS Scholarship for AI MSc (2023).
- Associate Research Fellow, QUB Momentum One Zero (formerly Global Innovation Institute), 2021–2024.

Professional Experience

Skills and Expertise

- **Econometrics as applied statistical science**: explicit assumptions, uncertainty quantification, and stress-tested inference for finance.
- **Financial statistical learning**: time-series ML, anomaly detection, and decision support in markets and trading settings.
- **Causal analysis for policy and practice**: treatment-effect estimation and causal ML for evaluation and decision-making under uncertainty.
- **Risk analytics**: tail risk, stress testing, and scenario analysis for financial institutions and portfolios.
- **Reproducible financial data science**: end-to-end pipelines (ETL), feature engineering, and governance for models used in operational contexts.
- **Responsible algorithmic methods**: interpretability, validation / controls, and attention to bias, fairness, and privacy where relevant.
- **Cross-disciplinary and industry collaboration**: translating research into practice with partners across finance, computing, and policy.